Spotter Full Stack Assessment — Weather-Aware Truck Routing

Nov 3, 2026 · @Ameema Rashid

Summary and links

A React + Django web app that compares 3 truck routes and recommends the safest, using the weather each truck meets at each checkpoint at its arrival time, the load weight and the travel time.

Deliverable	Link
Hosted application	spotter-routing-ameema.vercel.app
Backend API	spotter-routing-api.vercel.app (health: `/api/health/`)
GitHub repository	github.com/ameema28/spotter-routing
Loom walkthrough	loom.com/share/dd06cacf76684728bac43abc59615a0a

What the user does: enter an origin, destination, departure time and load weight, and pick checkpoint spacing of 10, 25 or 50 miles.

What the app returns: 3 route options on a map, the recommended route with a one-line reason, coloured checkpoints with ETA and risk level, route summary cards (distance, driving time, trip time with legal breaks, miles per risk level), and a weather heatmap along the corridor with a 0–48 hour forecast slider and truck positions.

How it works: routes, weather, risk and recommendation

All core rules are plain Python in `backend/routing/domain/`, written test-first.

Part	What the app does
Routes	Mapbox Directions returns at most 2 routes, so a third is generated (avoid-tolls plus interstate hub-city vias, in parallel). A candidate is kept only if ⩽1.25× the fastest time and ⩽75% shared road; the most distinct wins.

Part	What the app does
Truck ETA	65 mph cap plus US hours-of-service: 30-min break after 8 h driving, 10-h rest after 11 h (FMCSA summary and 49 CFR 395.3). Dallas → Chicago: 15 h driving becomes a 26 h trip.
Checkpoints	Every 10/25/50 miles plus the destination, each at its own ETA. Each checkpoint stands for the miles halfway to its neighbours, so Severe/High "miles" add up to the route length.
Weather	Open-Meteo hourly, batched: rain = rain + showers, snow = snowfall, wind = sustained 10 m speed. Rain/snow use the hour ending at or after the ETA; wind uses the nearest hour.
Risk bands	The brief's table with lower-inclusive edges (shared edges take the riskier band); 1.00 in/hr rain and 3.0 in/hr snow stay Severe; wind ≥55 mph is No Travel.
Combining + load	Worst of wind/rain/snow, then load rules that only raise the level: ≥55 mph No Travel; 45–54 mph and >30,000 lb No Travel; 35–44 mph and >40,000 lb Severe. Missing data is Unknown, never Low.
Ranking	Tiers: clean routes, then >25% unknown miles, then any No Travel. Within a tier, the brief's order: fewest Severe miles, fewest High miles, lowest average risk, shortest time. Only a clean route is recommended, otherwise "No safe route — consider delaying departure". The reason names the deciding rule.
Heatmap	≤250 grid cells around the routes, 49 hourly levels (hour 0 = departure) using the same rules. The slider only changes which hour is shown; no server call.

Architecture, performance and quality

Stack. React + TypeScript + Material UI + MapLibre (OpenFreeMap basemap, no key) on Vercel, calling Django + DRF on Vercel. One endpoint, `POST /api/route-plan/`, returns routes, checkpoints, breaks, summaries, ranking, the recommendation and the 48-hour

heatmap in about 11 KB gzipped. If the weather provider is busy, routes still load and the heatmap retries via `POST /api/heatmap/`, reusing cached cells.

Protecting speed and limits. Weather cached 30 minutes per point (snapped to 0.02°); per-client throttling of 10 searches a minute and 100 a day; clear errors (400, 422, 429, 502, 503, 504).

Measure	Result
Slider update (live, `?perf=1`)	About 2.6 ms React work; drawn in the next frame (15.6 ms median)
Lighthouse (live, desktop)	Performance 89 · Accessibility 100 · Best practices 100
Search time (live)	Fresh 3.3 s, repeat 2.1 s, cold start 7.7 s
Tests	Backend 465 (pytest), frontend 89 (Vitest)
Deliberate bugs injected	35, all caught; one gap found and closed
Heatmap vs checkpoints	95.9% agreement (117/122), rest one level apart
QA	Requirements checklist, 4 live searches, 28/28 UI scenarios pass (see `docs/QA.md`)

Decisions, limitations and AI use

Key decisions (full list with reasons in `DECISIONS.md`):

- Shared table edges go to the riskier band; every edge is a unit test.
- Worst of wind/rain/snow wins; load rules can only raise risk.
- Truck ETAs include federal hours-of-service breaks, so forecast hours match a legal trip.
- No Travel routes always rank last; if nothing is safe, recommend none.
- Heatmap built once per search; the slider changes only the colour rule, so it stays smooth.
- Free hosting on Vercel (no sleep, 300 s limit) with a heatmap retry when the weather service is busy.

Limitations and next steps:

- Car-profile routing with a speed cap; next, a truck routing API (height, weight, hazmat).
- Hours of service simplified (no 60/70-hour limits, sleeper berth, adverse-conditions extension, real rest-stop locations).

- Open-Meteo free tier and shared Vercel IPs can hit rate limits; next, a paid weather plan or fixed IP.
- Cache and throttling are per server instance; next, Redis.
- Misspelled places are matched silently (the matched name is shown); next, "did you mean".

AI assistance and time. I planned the work with Claude and built it with Claude Code, one block at a time in plan mode. I made every decision, reviewed each plan and the tests, and checked the results in the browser. Time spent: about 14.75 hours of the 16-hour limit, over 4 days (log in `PROGRESS.md`).